

FIRST BI-MONTHLY ISSUE—

A.N.C.

PLANET

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CARGO TO CALLISTO

by **JAY DREXEL**

Another famous

STAR-MOUSE story

by **FREDERIC BROWN**



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THE LAST TWO ALIVE!

A Novel of Exploding Suns by **ALFRED COPPEL**

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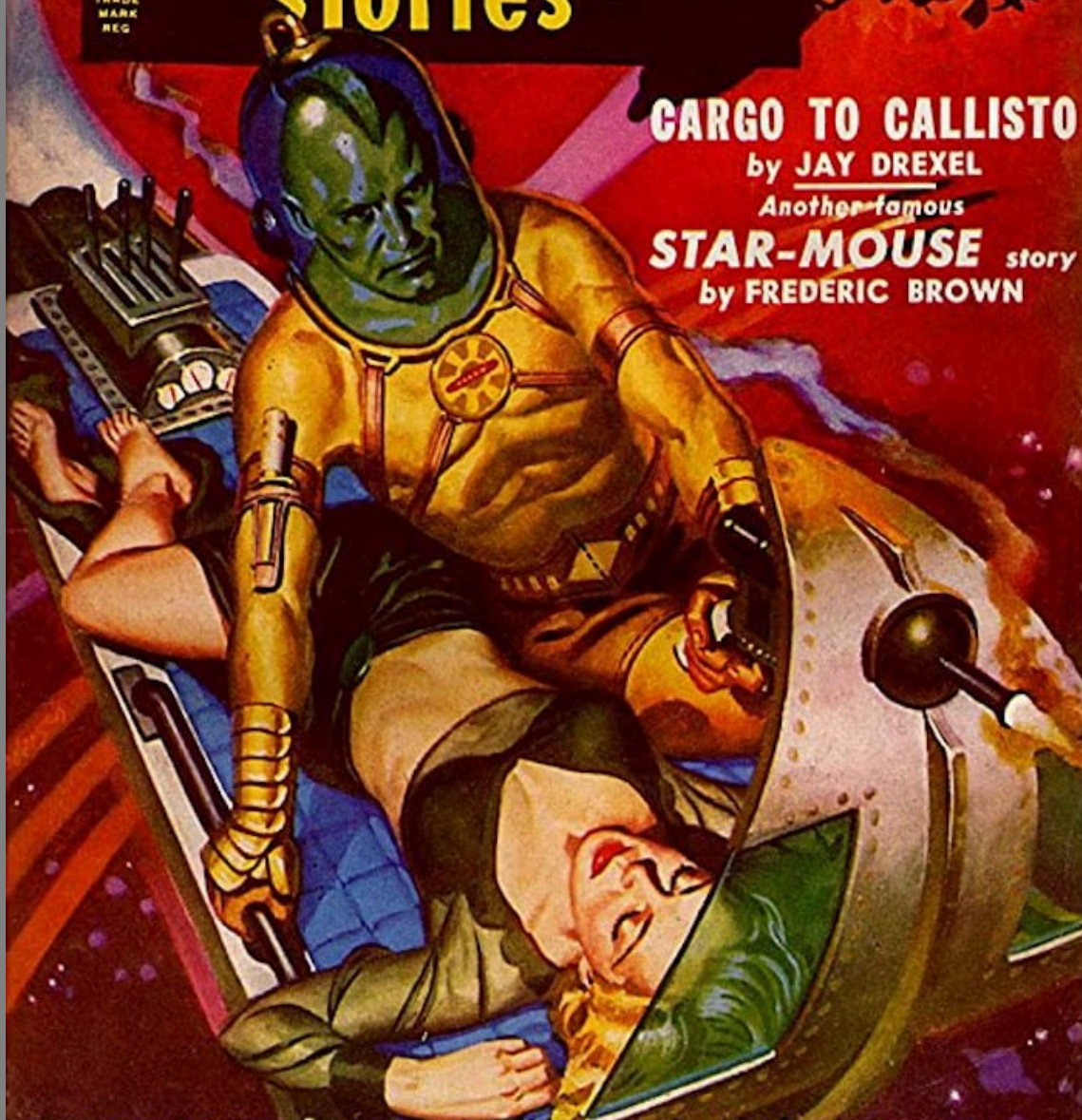
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*** START OF THE PROJECT GUTENBERG EBOOK MACHINE OF
KLAMUGRA ***

Machine of KLAMUGRA

By Allen K. Lang

**Captain Barnaby and Lieutenant Teajun stood
at the brink of that vast stone amphitheater,
staring wonderingly down at half-an-acre of
gadget. This glittering mass of million-year
clockwork was the Machine ... and soon it was
to judge them for their crime against Mars!**

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Klaggchallak, his fur nose-flaps pulled tight against his nostrils, stumbled up to the gleaming pinnacle of steel that seemed to offer shelter against the night. He felt a dust-storm gathering in the west, and knew that not even the tough skin of a Martian priest could withstand the angry whippings of sand lashed up by the wind-warlocks of the desert.

The old priest drew a tiny, folded *mal*-skin tent from his back-pack. Without haste, for he knew that the elder gods of Mars were watching his safety, Klaggchallak pitched the tent against the west stabilizer of the rocket, drawing the tough hide down to form a floor-flap and fastening it to the steel of the stabilizer with tough *mal*-hoof glue, which would hold fast in the fiercest winds of Mars. He looked for the sun and found it low in the evening sky, then crawled leisurely into the yurt, pulling the door-flap down after him and gluing it to the floor. He had for himself a secure cocoon into which the sand-devils could not force their probing fingers. Before he slept, the old priest fingered his beads, reciting his evening invocation to various benevolent and protective gods.

The falling sun threw a dancing star against the hull of the ship standing tall in its tail-chocks. A bewildering wail, the banshee-call of "Danger; ship jetting off!" sounded; but Klaggchallak slept on, hearing through his dreams only the howling of the wind.

Sixty seconds later, as prescribed in the General Regulations of the Extraterrestrial Service, a second sound began, that most fearful of noises, the sirening of the rocket exhaust. The Martian in his skin tent awakened and felt fear gnaw at his bones; fear induced by subsonic tremors from the rocket blast. Klaggchallak reached for his beads as the heat soaked into his thick, wrinkled skin.

In a moment the floor sagged beneath him. With the *mal*-skin pouch dangling ridiculously from its tail assembly, the EXTS rocket *Vulcan* rose with great gentleness from its tail-chocks, pushed up on its spraying jets.

Four seconds later the ship was a ruby flame above the low hills. Eight seconds later a charred bipod, a bifurcated cinder, tumbled down from space to strike near the jetoff field, where the *Vulcan*'s tail-chocks glowed dull red and the blackened ground smoldered. A moment later a bracelet of blast-

welded beads tumbled down from the sky, falling near the carbon hulk that a few seconds before had been Klaggchallak, a Martian priest great in wisdom and in honor among his people.

Captain Jan Barnaby and Lieutenant Kim Teajun of the Extraterrestrial Service stood before the Board of Inquiry at Denver, D. F. The President of the Board, a Chief Commander's star-on-silver gleaming at his right collar point, opened the proceedings:

"The military rocket *Vulcan*, EXTS light cruiser, is accused by the Martian authorities of causing the death of one Klaggchallak, a priest. They further claim that the death of Klaggchallak was caused by criminal negligence on the part of the pilot and co-pilot of the rocket, Captain Jan Barnaby and Lieutenant Kim Teajun, respectively. Such neglect being within the definition of murder in the Martian legal code, the Judging Authority of Mars demands that we deliver these two men to them for trial and eventual punishment."

The Commander stroked his grey hair thoughtfully as he looked up from his report to the two unhappy officers before him. "At ease, gentlemen." Barnaby and Teajun slumped. "While I'm inclined to agree with you two that Klaggchallak's frying was his own fool fault, I must say that you picked a damned poor time to become the instruments of his immolation. We had hoped to establish an extra-territoriality agreement with Mars, but the death of old Klaggchallak puts that out of the question. To further Martian tranquility, you men will have to return to Mars and face the Judging Authority there. If my feelings were all that is at stake, gentlemen, I'd tell the Marties to go trippingly to hell, and keep you here on Earth. But to do this would mean we'd be forced to abandon our bases and mines and surveys on all Mars. We'd be giving our European competitors a clear field." The Commander folded the report neatly, once and again. "Captain Barnaby and Lieutenant Kim, you'll be on the next Mars-ward ship. We can't help you if you're convicted by our fuzzy friends. You'll have to stay there and take whatever punishment they demand."

Kim was remembering a scene he and Barnaby had witnessed at Klamugra, the seat of the Martian Judging Authority. A Martian, convicted of murder, was being executed atop a high metal platform. A large portion of the city's population was gathered before the platform, watching the edifying spectacle of a fellow-Martian dying with horrifying slowness as the chocks of a vise pressed into his skull. They were bearcats for gladiatorial amusement.

"Do you gentlemen have any questions?"

Lieutenant Kim glanced at Captain Barnaby, then spoke. "Yes, sir. I'd like to know how long we're going to let the Marties push us around this way. Thirteen Martian priests are on our payroll, just because they demand it. We've got to stay five kilometers away from their cities, or pay a five-hundred credit fine. We can't spit without special permission from the Grand Council of Mars. We don't think like they do; why should we submit to being judged by their million-year-old laws? In all respect, sir, why does our Service act so weak?"

The Commander made a pyramid of thumbs and forefingers, and considered it. "Lieutenant Kim, I've been asking myself that question for the last ten years. We've had to pay tribute to gain the Marties' permission to stay on their god-forsaken planet. That tribute represents half the operating expense of the Martian Department of the Service, credits that should be spent on new ships and more men. We've behaved like a bunch of patsies ever since von Munger and Ley landed on Mars.

"Still, we're all soldiers, and we must follow regulations. We mustn't disturb the indigenous population on Mars; that's Regulation 'A-1.' If our policies grow distasteful to the Marties, they may call in the Europeans to take our place. We wouldn't like that. It's bad form to admit it, gentlemen, but I'm ashamed to give you this order. You're to jet off for Mars tomorrow morning; and on arrival at Klamugra, to deliver yourselves over to the Martian Judging Authority." The Commander rapped his gavel and stood; the two officers before the Board snapped to attention. "Board of Inquiry dismissed."

Fully aware that tomorrow's jettison would multiply by eight the hangovers they were breeding, Captain Barnaby and Lieutenant Kim sat that evening in the Denver Dive, alternating drinks of European vodka with rounds of California moon-dew. As Kim said: "Drink as much as you like, Barnaby; we're not driving in the morning."

"Tell me," Barnaby demanded of his co-pilot, "what you're thinking of, you Martie-roasting fiend of a Korean."

"I was considering the memory of the 'shlunk!' that Martian murderer's skull made when it finally gave in, that day at Klamugra. Do you remember, hard-headed Yankee?" Kim's eyes followed the blonde ecdysiast across the stage more from habit than present interest.

"Why did you have to remember that? 'Shlunk!'—ugh!"

"We're going to have to squirm out of this, Barnaby-*sunsang*," Kim said. "We'll have to beat that rap at Klamugra. It's not that I wish to avoid putting my head in a vise; it's only that it hurts me to see the Extraterrestrial Service made a monkey of this way. In a way it will even be a shame if we get off. Think of all the Marties who will miss the opportunity to see your punkin head smashed."

"You orientals have noble souls, Kim."

The blonde stripper, having uncovered as much of herself as she could without resorting to dissection, jumped down from the stage and walked over to the two EXTS officers. "Would you gentlemen like to buy me a drink?" she asked.

Kim's eyes roved abroad in a brief anatomy lesson, but Barnaby said, "I'll buy you one a couple of weeks from now, if I'm not laid up somewhere with a splitting headache." He stood unsteadily and tossed a ten-credit certificate on the table. "If you're really thirsty, get a drink out of that."

Kim reluctantly followed his superior officer from the bar. At the door he turned and called back to the blonde, "Don't catch cold, child. I'll be back."

The dawn jettie was miserable, as jetties always are. Four days brought the ship within falling-distance of Mars; soon the jets thundered as it backed into a pocket of hills outside Klamagra. The air-pumps hammered to bring the air pressure inside the hull gradually down to that of the outside, so that instruments and equipment wouldn't be subjected to a sudden lowering of pressure. The men inside the ship slipped plastic helmets over their heads, checked the tiny air-pumps on their shoulders, and drew on heavy gloves and boots.

When the port swung open Kim and Barnaby climbed down the ladder to the blast-blackened sand. The sergeant of EXTS Provost Marshall who had accompanied them walked with the officers to a hill overlooking the ancient Martian city of Klamagra, which stood on a terrace about five kilometers to the north. The red adobe walls of the city, testimony of the ancient days when Mars had enough water to allow its use for brick-making, blended with the distance to seem a part of the red desert sand.

A cloud of steam and dust appeared between the hill where they stood and the city. Captain Barnaby un-leathered his binoculars and pressed them to the eyepieces of his helmet, and made out a hopping jeep, its top enclosed in plastic and a trio of supercharger coils poking through the sides of the hood. Clouds of steam followed the jeep as its exhaust streamed out into the chilly air.

In a moment the jeep spun up the hill and ground to a halt. There was a pause as the men inside the jeep fitted their helmets on their shoulders, checked their air-pumps, and drew on their gauntlets. Then the plastic bubble lifted back, a sergeant jumped out from under the steering wheel and saluted, and a Colonel, EXTS Intelligence, walked up to Captain Barnaby and Lieutenant Kim. "Gentlemen," he said, "I'm Colonel Lee Montgomery, Commanding Officer, Third Sector. It is my unpleasant duty to turn you over to the Chief Technician of the Martian Judging Authority, who is Rhinklav'n, here."

At these words a tall Martian unfolded himself from the back seat of the jeep. He climbed out and bowed before Captain Barnaby. "I am Rhinklav'n,

Captain." The thick fur nose-flaps, looking like ear-muffs pulled across his muzzle, muffled Rhinklav'n's high-pitched voice so that it gave the effect of coming from the bottom of a rain barrel. "You are to accompany me to Klamugra to be judged by the Machine, of which I am the Honored First Technician."

Barnaby and Kim bowed slightly to acknowledge Rhinklav'n, then crawled into the back seat of the jeep, next to Colonel Montgomery.

Rhinklav'n and the sergeant sat up front. The sergeant pushed a button on the instrument panel, and the plastic top of the jeep dropped down to cover them. As the engine started, the jeep's air-pump drew in air until the atmosphere was thick enough for human lungs. The Martian squirmed uncomfortably in the heavy air while his human companions threw off their helmets. Lieutenant Kim gratefully drew a deep breath of air, and regretted it immediately. What with the million-year water shortage the Martians had lost even the word for bath. Besides, the most popular article of Martian cuisine is a bulb strikingly similar to the terrestrial garlic plant. Captain Barnaby turned to Kim. "Mars has a distinguished atmosphere, hasn't it?" He spoke in English, rather than in the Esperanto lingua-franca of space.

"Indeed it has," Kim agreed. "What was old fuzz-face up there talking about when he spoke of 'the Machine,' Colonel?"

"The law of Mars is the most rigidly systematized in the solar system," Colonel Montgomery replied. "Several millions of years ago, a bright Martie got the idea that it was unwise to trust mortal judges with a problem so important as the sentencing of criminals. So he called in a lot of mechanics—ancient Mars had some pretty fair engineers, though they never discovered electricity—and had them build a judging-machine. Since the climate is right and the machine was built of a stainless steel, it's still here and still being used. It's an enormous thing; spreads over half-an-acre in a big amphitheatre in the center of town. It's an analogue computer, rather clumsy by terrestrial standards, but nevertheless well-built. You know the principles of analogue calculators. Instead of working with coded, position-valued impulses, like the electronic astrogator on our rockets, the mechanical machines solve problems by making use of the physical analogies between cogs and gears and differentials."

"Do you mean that we're going to be punished or set free by a bunch of clockwork, colonel?" Kim asked.

"In a way, yes. The Machine is a most impersonal judge. That fact won't help you, though. Martian legal code is strict about killing, there being some thirty-odd degrees of murder, ranging in seriousness from a 'simple homicide to secure a mate,' the punishment for which is death by dehydration, most often; to 'killing to secure for oneself material benefits,' for which there exist more subtle forms of death by torture."

"Like getting a small-head-size in a vise?" Captain Barnaby grunted.

"That's the usual punishment for murder in the seventeenth degree, where the crime is usually 'killing for spiritual advancement.' You see, each crime is given special study by the Machine. A great many factors are fed in, collated with certain constants within the Machine, processed through several dozen stages, and finally combined into a single number, which represents the punishment called for. By the way," the colonel studied the back of Rhinklav'n's head, "no consideration of the truthfulness of the 'defendant' is entered into the Machine. It is presumed that should a man say that he did not commit a crime, he didn't; if he did, he'd admit it. Martians have a peculiar character defect that prevents them from lying."

"A defect from which we humans are fortunately free," Kim grinned.

"That's no out," Colonel Montgomery countered. "They have witnesses who saw Klaggchallak fry. Besides, we prefer to have the mass of Marties ignorant of the average earthling's penchant for prevarication. It saves the Service a lot of money not to have to prove anything it tells our hairy hosts out here."

The jeep hit the first of the series of low terraces which set the city of Klamugra up from the surrounding desert plains, and the little car bounced high off the sand. Colonel Montgomery looked startled, as though he'd just remembered something. "You know, in my ethnological fervor I didn't realize what you two men are in for. Cosmos! I'm practically delivering you up as human sacrifices!"

"We came to that conclusion five days ago, colonel," Lieutenant Kim dryly observed.

"I can see what the Fleet Commander meant when he said that he was giving me 'a most unpleasant assignment.' Hell, I don't think the Machine is able to give a judgment of 'not guilty'." Colonel Montgomery gazed toward the city they were approaching. "We've got to turn you in. We can't risk a blowup with the Martian Grand Council. There are rumors that ..." the colonel glanced again with suspicion at the back of Rhinklav'n's hairy neck, as though suspecting that the Martian might be able to puzzle out the meaning of their conversation, though it was in English. "There are rumors that the -artiansMay have an agent among the -ussiansRay. We can't risk having the borsch-eaters more popular out here than the Western Powers." The jeep bounded up the last of the terraces and through an opening in the city wall. The adobe buildings raced past, and with a final bound the jeep came to the edge of the huge, circular bowl which held the Machine.

"There's your judge," Colonel Montgomery said, speaking in Esperanto again. "I haven't much hope to offer you. For one thing, you're the first humans ever to be judged by the Machine."

The men picked up their helmets and air-pumps and adjusted them on their shoulders. Rhinklav'n drew his furry nostril-flaps down into place against the sudden change in pressure. The plastic top of the jeep flew back on its springs and the men climbed out, stretching their cramped muscles. The radiophones in the helmets buzzed, and the colonel gave Captain Barnaby a last word. "I want to impress you with the fact that the Service cannot protect you, from this moment onward. If you escape being killed it must be on your own merits. And don't start shooting Marties—won't do you a bit of good. There's a lot at stake for Earth here. Good luck, men!" Colonel Montgomery saluted, and he and his sergeant jumped back in the jeep, slammed the top down, and whirled away.

Rhinklav'n turned to the two EXTS officers. "Gentlemen, I've assigned you quarters here, near the Machine. Will you follow me?" Kim and Barnaby followed the Martian a short distance from the edge of the amphitheater to a lone adobe building, one story high and about ten meters square. "Here are your quarters, where you'll stay tonight. Your judging is set for tomorrow morning."

Captain Barnaby glanced into the building and was surprised to see that it closed with an airlock, had terrestrial canned foods on neat shelves, and had regular Service cots in place of the rough *mal*-leather mats that the Martians slept on. "It was good of you to go to all this trouble just for myself and Lieutenant Kim, Rhinklav'n," the Captain said.

The Martian paused at the door. "It's not just for you, Captain. Five other terrestrials have committed crimes of various proportions within the last few weeks. They will also be tried here, after your case is disposed of." Rhinklav'n left, considerately closing the airlock door and starting the pump on his way out.

Lieutenant Kim took a can of "B" ration beans down from the shelf and thoughtfully began to open it with his Service knife. "Captain," he said, "this sort of thing could drive our Service from Mars. If the Marties consider it their right to judge every Earthling who runs a jeep into a farmer's *mal* or lands half a meter too near one of their cities, we won't have a man on the planet in a couple of years."

"Kim, we're precedents."

"What do you mean, Yankee?"

"If the Marties succeed in convicting us of murder in some unheard-of degree by using that overgrown Erector Set of theirs, we'll be only the first two of a long string of EXTServicemen to be executed under Mars law. We can't let them do it." Captain Barnaby paused a moment to pour himself out a plateful of beans. "Kim, what was that process you used to rely on back in EXTS Academy in Denver? The one that gave you the right answers after you found that your first solutions to our astrogation problems were a few hundred thousand kilometers off?"

Kim stopped chewing for a moment in surprise. "You mean that you got through the Academy without using the 'finagle factor'? No wonder you made captain so soon. It's simple: I'd look up the right answer in the Service charts, find by what factor my solution was off, and introduce that factor into my next calculation, making it inconspicuous under a lot of mathematical camouflage. Don't bawl me out about it, Barny; I just couldn't see letting my extracurricular activities suffer for my schoolwork."

"Yes, you did a lot of your studying at the Denver Dive. No matter, little man. Eat hearty and get some sleep." Barnaby stirred his beans thoughtfully. "We've got a big day ahead of us tomorrow."

Early the next morning a subordinate technician of the Machine hammered on the airlock. The two terrestrials pulled on their heavy jackets, fur boots, and gauntlets, started the little air-pumps on their shoulders, and opened the lock. "The honored First Technician of the Machine invites your presence at your trial, which is to begin very soon," the Martian said, speaking halting Esperanto. Kim and Barnaby followed him to the edge of the Machine bowl. There had been several changes made during the night. An elevated platform had been set up, identical to the one used in the bloody execution they'd witnessed. About twenty Martians were clustered around the Machine, some of them making last-minute adjustments in the mechanism; others, evidently sightseers, gazing curiously at the two principals in the trial.

Rhinklav'n was waiting, his nose-flaps drawn over his nostrils to keep the cold morning air from cutting into his lungs. "I am pleased that you come," he said. "The Machine is fully assembled for your problem." He pointed down toward the Machine, a vast cluster of separate stages connected by rods. "On the far right, in that small building, is the power source of the Machine, a mercury-turbine engine. We can't spare the water to make steam, you know. The first stage contains the Martian actuarial tables, the second has the actuarial system for determining the probable life-spans of you two Earthlings. That's without taking into consideration the probability that you two will be executed as a result of the judgement of the Machine."

Lieutenant Kim nodded. "Most ingenious. But I'm afraid that there's a factor that you've omitted."

"We've made no factual error, Lieutenant," Rhinklav'n insisted. "The value of the sage Klaggchallak is represented there—" he pointed to the fourth stage, "and your social value to the people of Mars is here represented." Rhinklav'n waved one mitten-like hand toward the fifth stage. "If you'll examine that stage, you'll observe that your value is negative: the shaft

representing it revolves in a direction opposite to that of the others. Yes, you'll surely be executed."

Captain Barnaby nodded, as though the reiteration of the probability of his early demise troubled him less than the philosophical question he'd stumbled across. "Still, as my subordinate officer has said, there's a factor which you seem to have omitted. In the terminology of terrestrial psychomentering, this quantity is called—what did you say it was, Lieutenant Kim?"

"The 'finagle factor,' sir."

"Really?" Rhinklav'n asked, the light of scientific inquiry in his eyes. "I thought I'd taken all the variables of Earthling physiology and psychology into consideration when I set up the plans for the trial. What are the mathematics of this 'finagle factor'?"

Captain Barnaby put one foot up on a connecting shaft, as though he were in the Denver Dive, discussing the relative merits of two video dancers. "As you've doubtless noticed in your extensive study of the Terrestrial mind, Sir Honored First Technician of the Machine, we of Earth place almost equal value on a man's intelligence and on his financial standing as criteria of his worth."

"Yes," Rhinklav'n mused, "I noticed your preoccupation with both intelligence (a minor mental quality, by the way; far inferior to spiritual insight or time-sense) and with the individual's possession of Western Credits."

"As I said," Captain Barnaby continued, "there exists a precise formula, developed by the...."

"By the Noyoudont Dentifrice Laboratories," Kim supplied.

"Yes; their laboratories developed the mathematics of the finagle factor. Briefly, it is this: the square root of the product of Intelligence Quotient over one hundred times the number of credits the individual has

outstanding. Or, written algebraically:" Barnaby knelt down and traced in the sand with his gloved index finger:

$\sqrt{(\text{IQ}/100 \times \text{money in the bank})}$

"Quite a simple equation, easily represented on the Machine," Rhinklav'n observed. He called a subordinate technician to his side and spoke to him in the clicking polysyllables of the Martian language. Turning again to Captain Barnaby, he asked, "And what are the values of 'IQ' and 'Money in the Bank' for you and Lieutenant Kim?"

"Our combined IQs total about 243. How many credits do you own, Lieutenant Kim?"

"Hell, sir; I've got more debts than credits."

"Figure up your debts then, Lieutenant."

Kim raised his right gauntlet, drew a pad of paper and a pencil from a pocket at the back of his hand, and scribbled rapidly. "If we get out of this, Captain, I'll owe about 1046 credits. Subtracting pay due for the last semi-annual period, I owe 437 credits."

"And I have debts totaling 600 credits," Captain Barnaby said thoughtfully. He turned to Rhinklav'n. "The debts of myself and Lieutenant Kim, Sir Honored First Technician of the Machine, total 1037 Western Credits. Being debt, that's a negative number, of course."

"Of course, Captain," Rhinklav'n agreed. "The Machine can handle any sort of number, even a negative number. You noticed that your social value to Mars was easily represented as a minus-number." Rhinklav'n talked rapidly to his assistant and handed him the values of the finagle factor, rewritten in Martian ideographs. He faced Captain Barnaby again. "It will take us about an hour to enter this new factor into the Machine," he said. "You'll not mind waiting?"

"No, not at all," Barnaby murmured. He and Kim leaned against the inside wall of the amphitheater, watching the Martian technicians hurry about; they removed gears and replaced them with gears of another ratio; they connected a stage consisting of eccentric cams strung on shafts; and they installed a mass of machinery at the sixth stage, where the operation of extracting square root was to take place. Kim, comparing the heavy gears

and levers of the Machine with the compact tubes of his electronic astrogator, remarked, "It's like using a trip hammer to crack a walnut."

After a few minutes of watching, Kim and Barnaby became conscious of an intruder within their helmets, a most unpleasant odor. They glanced up to the edge of the bowl. The Martian sightseers were sitting up there, dangling their legs above the Machine and utilizing the pause in the proceedings to eat their picnic lunches. They were busily unwrapping bundles of food from the *mal*-skin pouches hanging by their sides and eating as they watched the technicians work over the Machine.

One of the tourists, judging from his height a young male, threw a small parcel toward Kim. The lieutenant picked it up and unwrapped it. The stench of Martian garlic became unbearable as Kim stared at the unidentifiable tidbit of meat the Martian had thrown him; the air-pump on his shoulder drew the redolence into his helmet in such quantities that Kim's eyes burned. He gestured to show that, while his every instinct demanded that he eat the delicious morsel, he couldn't take his helmet off to do so. With an elaborate pantomiming of sorrow, Kim pitched the gift back up to the Martian boy.

A few adjustments later the technicians filed up from the Machine pit. Rhinklav'n walked over to the two EXTS officers. "If you gentlemen will accompany me, we'll begin the trial at once."

Kim and Barnaby walked together up the steps that led from the Machine, then turned and looked down at the dozens of stages of complex machinery, into which memory and intelligence of a sort had been built. Rhinklav'n pointed toward the fifth and sixth stages. "It is there that the combined finagle factors of you men will be calculated. The fifth stage is quite simple; it will perform the necessary division and multiplication. The sixth stage will extract the square root of the product derived by the fifth. The next six stages of machinery contain the variables of terrestrial behavior, which I and my colleagues calculated from Earth texts. The other stages on the field, fifty-three of them, will collate the results of the calculations of the first twelve stages with our legal code and determine punishment. The final

product will appear at the sixty-seventh stage, represented as the speed of rotation of a single shaft. The revolutions-per-time-interval are decoded by a simple formula to determine the punishment to be levied upon you. Doubtless, it will be some unpleasant form of death."

Kim muttered that he wished that Martians had a bit more tact.

Rhinklav'n waved a hairy arm toward his assistant who had remained below in the Machine pit; and that Martian ran to the power house to start the mercury-turbine engine that ran the Machine. With a whistling that set the thin atmosphere trembling for miles around, the turbine began to turn.

The sightseers on the edge of the amphitheater wrapped up the scraps of their lunches, replaced them in their *mal*-skin picnic hampers, and stood up to watch the Machine. Kim and Barnaby paced up and down along the edge of the bowl, looking down upon the mechanical cerebation being performed by the huge Machine. With a smooth transfer of power from one stage to the next, the first problem—the probable duration of Klaggchallak's life when it had been interrupted by the jets of the *Vulcan*—was solved, and the mechanism of the second stage began to revolve.

"They're seeing how long we can be expected to live, now," Captain Barnaby commented.

That problem fled through a mass of gears and cams; and the partial solution, the sum of the two earthling's life expectancies divided by that of the priest Klaggchallak, ran across a shaft to the third stage, which would determine the old priest's value to the society of Mars. On into the fourth stage the problem flowed, to combine all previous factors with the earthlings' social value to Mars, a negative number.

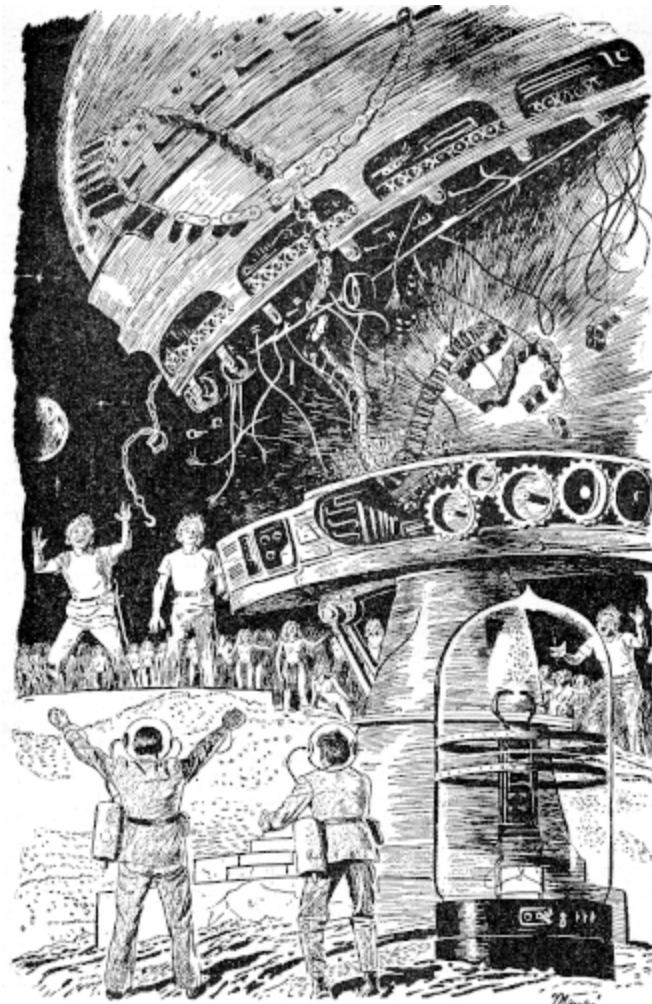
In a few moments the problem had progressed to the fifth stage of the Machine, where the first steps of the 'finagle factor' were solved. The product, a negative number as could be seen by the reversed rotation of the main shaft, bowed into the sixth stage, which was to extract square root.

The turbine howled protest as it was forced to overcome the inertia of the sixth stage; but a governor at the input stage held the shaft-speed constant.

The seventh stage, all ready for the problem when it should appear from the sixth, held all the computations of the first four stages in its smoothly-turning entrails. The initial portion of the sixth stage began to move slowly.

There was a sudden, grating noise as the feed-in gear of the fifth stage came in contact with a solution gear of the sixth which refused to move. The whine of the mercury-turbine engine was shaking the ground beneath the two officers' feet now.

As the Martian technicians and picnickers looked on in amazement, the shaft between the fifth stage and the sixth began to twist like a stick of moist putty. The sixth stage strained and shuddered, then followed the twisting shaft over, tearing its moorings from the ground and smashing upside-down. The seventh stage entered into the chaos, ripping out anchors of steel-in-concrete and slamming onto its side. In a moment all the machines in the bowl were muttering and straining against the earth. Rhinklav'n ran to the stairway that led down into the pit. In the adobe powerhouse the mercury engine was whirling at twenty times its optimum rate, tearing the atmosphere with the sound of its screaming power. There was the rattle of shrapnel exploding within the walls of the powerhouse as the turbine threw off the restraintment of its governor. The whole field within the bowl was a mass of twitching clockwork, shaken by the final stormings of the suicidal turbine. Bars of shining steel twisted and snapped, gear teeth flew singing through the thin air. The final chain of stages tore itself loose from anchoring and crashed to its side. There was a final roar of defiance from the turbine, and the powerhouse walls dissolved before an out-rushing blast of superheated mercury. Kim and Barnaby threw themselves to the ground as the din increased for a moment, and the Martian sightseers sought refuge behind nearby buildings. Suddenly, the Machine was silent, except for the tinkle of scraps of metal falling to the cement.



Bars of shining steel twisted and snapped, gear teeth flew singing through the thin air....

"Looks as though we were too much for judge, jury, and D.A.," Kim murmured into his radiophone. Barnaby nodded, then cautiously climbed to his feet.

Rhinklav'n climbed back up the stairway to the brink of the amphitheater-become-junkyard. He shoved his way through the questioning crowd of Martian sightseers without a word. "Looks like he's going to cry," Lieutenant Kim commented into his radiophone. True, Rhinklav'n's nose-flaps were hanging limply down below his chin, a sure sign of great emotion in a Martian.

Rhinklav'n faced Captain Barnaby wordlessly for a moment. "You may leave now," he said at last. The Martian turned his back on the captain to look down again on the wreck that had been his beloved Machine.

The two EXTS officers wandered about Klamugra, the cynosure of all Martian eyes, though no one tried to stop them or ask them questions. Lieutenant Kim finally spotted a radio tower jutting up above the red adobe buildings. Hurrying in the direction of the tower, Kim and Barnaby found the Klamugra headquarters of the Extraterrestrial Service.

Colonel Montgomery jumped to his feet as they came in, a look of bald disbelief on his face. "Man, I'm glad to see you two! I was about to storm out like a knight in shiny armor and save you from the Marties." He waved his hand toward the helmet and rifle lying on his typewriter table—"If I'd gotten there too late, I'd have ruptured interplanetary friendship for sure!"—and indicated a decanter on his desk. "Have some: that's Edinbrough scotch, not Los Angeles moon-dew. Tell me why I happen to be talking to you now instead of making up a couple of packages for your next-of-kin."

"We wrecked their damn Machine," Kim said happily, dropping his helmet and gauntlets to the floor and measuring out several fingers of the colonel's scotch into his ration can.

"To be a bit more accurate," Captain Barnaby corrected, "we drove the Machine insane." He poured himself a stiff shot of scotch and downed it with appreciation.

"Our personalities are so complex that the Machine blew up all over the landscape when it tried to understand them," Kim said. He dragged a chair out from behind the typewriter table and sat down, carefully balancing the ration can.

"It's rather as though we should set our electronic astrogator to work on a problem with three variables in five dimensions, rather than in four," Captain Barnaby explained. "As you told us, the Machine was a mechanical-analogus calculator. It can multiply, divide, add, square and

cube, and extract roots. It performs these operations by coding numbers into mechanical relationships."

"Just a big adding machine," Kim commented irreverently.

"And our 'finagle factor' was too much for a mechanical system." Captain Barnaby briefly explained to the colonel how he and Kim had induced Rhinklav'n to add their invented factor to the Machine's setup. "You see, the finagle factor resolved itself into the square root of a negative number. An electronic calculator, like our astrogator, could extract the root of a minus-number: 'imaginary' numbers of this sort are implicit in its circuit. The Martian Machine out there couldn't do this though. Since there is no mechanical analogue for an imaginary number, the Machine tried to extract the square root of our finagle factor in the same manner in which it would attempt to extract the root of a real number."

Kim drained, his ration can neatly and remarked, "The Machine couldn't do what it had to. All the power of the turbine was thrown into the root extracting system, which wouldn't revolve. So the Machine went nuts, pardon me, sir, and blew its top. Wrecked the power source and all sixty-seven stages. With the square root of minus one, we busted up a Machine half a million years old."

"What now?" Colonel Montgomery asked, rhetorically.

Captain Barnaby studied the bottom of his ration can a moment. "Well, sir, Rhinklav'n was more puzzled, than angered. He wanted to judge humans not out of malice, but from a genuine scientific curiosity. He wanted to see how the Machine would act with an alien problem. His Machine is too badly broken-up ever to repair. He'll have to find another method of judging criminals, first of all. Martian society is founded on strict law."

"Just a moment." The colonel got up from his desk and went down the hall to a door marked "Judge Advocate General's Department, EXTS." He returned with a heavy book, bound between khaki-board covers. "We'll give this book to Rhinklav'n, and you gentlemen may return to the Denver Joint."

"Dive, sir," Kim corrected.

"Yes, Lieutenant." Colonel Montgomery handed the big book to Captain Barnaby. "Take this to Rhinklav'n before you leave, Captain."

Barnaby turned to the title page and read in Esperanto, "Blackstone. *On the Study of Law.*"

*** END OF THE PROJECT GUTENBERG EBOOK MACHINE OF
KLAMUGRA ***

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